

Report
Of
SQL and Database Management Systems
(AICTE – EduSkills Virtual Internship)

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.
In partial fulfilment of the requirement of degree of
B.Tech CSE



By
Roll Number of Student: CS-2341315
Name of Student: PRATEEK SINGH
Batch: B.Tech CSE AIML & 7th Semester

2026-27

CANDIDATE'S DECLARATION

I, Prateek Singh do hereby declare that the internship report titled "**SQL and Database Management Systems**" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Prateek Singh

Roll No.: CS-2341315

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful to AICTE and EduSkills Academy for giving me the opportunity to undertake this virtual internship, and to the EduSkills mentors and programme team for their guidance, invaluable constructive criticism and friendly advice during this work. I am also thankful to the School of Computer Science and Engineering, IILM University, for their encouragement and support.

I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

Student Name: Prateek Singh

Roll No.: CS-2341315

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INTERNSHIP COMPLETION CERTIFICATE

CERTIFICATE
OF INTERNSHIP







PRESENTED TO

Prateek Singh

has successfully achieved the Internship Credential by completing a 8-week program in the domain of

SQL और Database Management Systems

During the internship, the learner demonstrated proficiency in:

- Database Fundamentals और Course Kick-off & Relational Database Design और SQL Basics
- Advanced SQL Queries और Data Aggregation & NoSQL Databases - Concepts और MongoDB Basics
- Data Aggregation और ETL SQL के साथ & Power BI के साथ Data Visualization - Fundamentals
- Power BI में Data Modeling और Relationships & Databases को Full-Stack Applications (Node.js & Express) के साथ एकीकृत करना
- Databases को Full-Stack Applications (Python & FastAPI) के साथ एकीकृत करना & Data Integration और API Consumption
- Database Performance Optimization & Database Security और Backup Strategies
- Scalability और High Availability Concepts & Real-World Case Studies और Best Practices
- Capstone Project: एक Data-Driven Business Analytics Dashboard बनाना & Advanced NoSQL Querying और Use Cases (MongoDB)
- Capstone Project: एक Data-Driven Business Analytics Dashboard बनाना
- Capstone Project: एक Data-Driven Business Analytics Dashboard बनाना

This certification recognizes the successful completion of the internship requirements.



Issued on: Sep 10, 2026
Certificate ID: 2026-D6F6DF349A



Milu Swain
General Manager
(Learning & Development)
EduSkills

PROJECT DESCRIPTION

4.1 Introduction

Data is at the core of every modern software system, and the ability to design, store, query, secure and analyse it is a fundamental skill for a computer science graduate. Internships bridge the gap between academic knowledge and real-world applications by giving hands-on exposure to industry-relevant tools and practices.

This report presents the 8-week AICTE – EduSkills Virtual Internship in the domain of SQL and Database Management Systems. The internship covered relational database design and SQL, NoSQL databases (MongoDB), data aggregation and ETL, data visualization with Power BI, integration of databases with full-stack applications (Node.js & Express and Python & FastAPI), database performance, security and backup, scalability and high availability, and a capstone project on building a data-driven business analytics dashboard.

4.2 Organization Profile

Organization Name: EduSkills Academy (EduSkills – "Nation Building Through Skills")

Domain: EdTech & Skill Development

Services: Virtual Internships and industry-oriented skill programmes

Supported by: AICTE (All India Council for Technical Education), under the patronage of the National Internship Portal

Recognition: ISO-certified organization and Great Place To Work certified

Internship Overview

- Internship Mode: Online / Virtual
- Domain: SQL and Database Management Systems
- Duration: 8 Weeks (April 2026 – June 2026)
- Platform Used: EduSkills virtual internship platform (AICTE – EduSkills)
- Role: Intern
- Offer Letter Ref. No.: C16Y2026M051441098
- AICTE Student ID: STU68346ab0a07001748265648

4.3 Problem Statement

Modern organizations generate large volumes of data from sales, customers, applications and web services. To make use of this data, it must be stored in well-designed databases, retrieved efficiently, protected against misuse and loss, integrated with applications through APIs, and finally presented as meaningful insights for decision-making. Students often learn these topics in isolation, without seeing how a relational database, a NoSQL store, an ETL process, a backend API and a reporting tool work together.

This internship addresses that gap through a structured, week-wise programme that moves from database fundamentals to advanced querying, data integration, performance, security and scalability, and ends with a capstone project: building a data-driven business analytics dashboard.

4.4 Project Objectives

- To understand database concepts and design normalized relational databases using ER modeling and SQL (DDL and DML).
- To write advanced SQL queries using joins, subqueries, aggregation, GROUP BY and HAVING.
- To understand NoSQL systems and perform CRUD and aggregation operations in MongoDB.
- To apply ETL and data transformation techniques and visualize data using Power BI (data modeling, relationships and basic DAX).
- To integrate databases with full-stack applications and build RESTful CRUD APIs using Node.js & Express and Python FastAPI with SQLAlchemy.
- To apply performance optimization, security (roles, permissions, SQL injection prevention) and backup strategies.
- To understand scalability and high-availability concepts such as replication, sharding and clustering.
- To build a data-driven business analytics dashboard as the capstone project.

4.5 Scope of the Project

The scope of the internship covers the complete data lifecycle: designing the database, loading and transforming data, exposing it through APIs, securing and tuning it, and

presenting it in an analytics dashboard. The learning can be extended in the future in the following directions:

- Working with larger, real-time datasets and cloud-hosted database services.
- Adding automated ETL pipelines and scheduled data refresh for the dashboard.
- Implementing replication, sharding and automated backup in a production-like environment.
- Extending the analytics with advanced DAX measures, time-series forecasting and geospatial analysis.

4.6 Technologies and Tools Used

Relational Databases: SQL, PostgreSQL – DDL, DML, joins, subqueries, aggregation, indexing, roles and permissions.

NoSQL Database: MongoDB – document model, CRUD operations, aggregation pipeline, advanced and geospatial indexing, time series data modeling.

Data Analytics & Visualization: ETL techniques, Power BI (data connections, model view, relationships, basic DAX, dashboards).

Backend & Integration: Node.js and Express; Python with FastAPI and SQLAlchemy; RESTful CRUD APIs; API consumption using the requests library.

Concepts: ER modeling, normalization, query optimization, SQL injection prevention, backup and recovery, replication, sharding, clustering, high availability.

4.7 System Architecture

The capstone solution follows a layered architecture. Data from business datasets and API sources is prepared through ETL, stored in relational (PostgreSQL) and NoSQL (MongoDB) databases, exposed through REST APIs built with Node.js/Express and FastAPI, and finally modelled and visualized in Power BI as a business analytics dashboard.

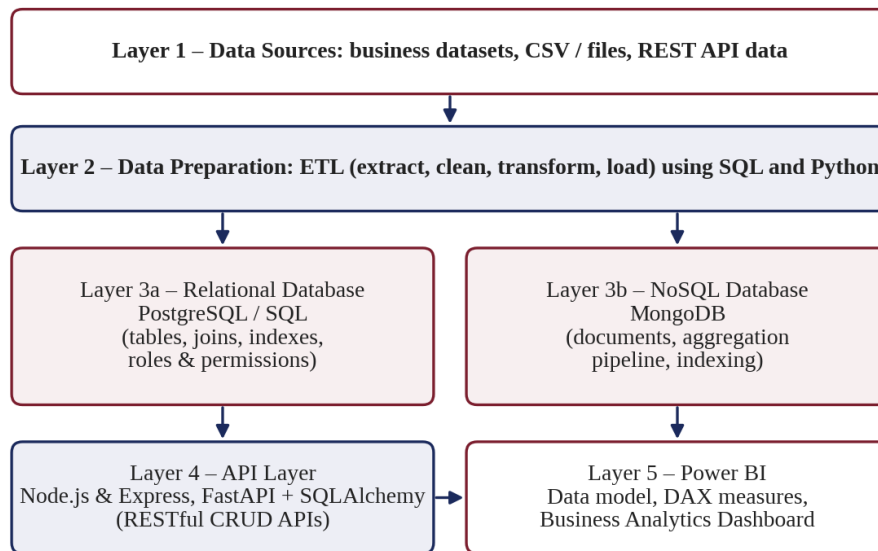


Figure 4.1: Layered architecture of the data-driven analytics solution

4.8 Methodology

The internship followed a week-wise structured learning plan. Each week combined concept learning with practical exposure, weekly assessments were used to track progress, certain modules required project documentation and assignments, and the final week included a Final Assessment Test. The final grade was based on overall performance across weekly assessments, project submissions and the final test. Table 4.1 summarizes the plan.

Table 4.1: Week-wise structure of the internship

Week	Module	Description
1	Database Foundations & Relational Design	Introduction to database concepts, SQL vs NoSQL systems, environment setup, relational database design, ER modeling, normalization, DDL and basic DML operations (SELECT, INSERT, UPDATE, DELETE).
2	Advanced SQL & NoSQL Fundamentals (MongoDB)	Advanced SQL queries including joins, subqueries, aggregation, GROUP BY/HAVING; introduction to NoSQL concepts, MongoDB document model, and CRUD operations.
3	Data Aggregation, ETL & Power BI Fundamentals	Advanced SQL aggregation, ETL concepts, data transformation techniques, introduction to the Power BI interface, data connections, and building basic dashboards and visualizations.
4	Data Modeling & Full-Stack Database Integration (Node.js)	Power BI model view, relationships, basic DAX, performance optimization; integrating databases with full-stack applications using Node.js & Express, PostgreSQL connection, and CRUD API development.

Week	Module	Description
5	Backend APIs & Data Integration (FastAPI)	FastAPI with SQLAlchemy, building RESTful CRUD APIs, API consumption using the requests library, storing API data, and combining multiple datasets.
6	Database Performance, Security & Backup	Performance metrics, indexing strategies, query optimization, configuration tuning, user roles & permissions, SQL injection prevention, backup and recovery strategies.
7	Scalability, High Availability & Real-World Use Cases	Scaling strategies, replication, sharding, clustering concepts, case studies from e-commerce, social media and financial services, and consolidated best practices.
8	Advanced NoSQL, Time Series Data Modeling & Capstone Project	MongoDB aggregation pipeline, advanced indexing, geospatial queries, time series data modeling, data quality, reproducibility, model validation and deployment concepts; Capstone Project and Internship Grade Point Assessment.

Capstone Project: Data-Driven Business Analytics Dashboard

The capstone brought the individual modules together. Business data was extracted and cleaned using SQL and ETL techniques, stored in databases, combined with other datasets (including data consumed from APIs), and loaded into Power BI. A data model with relationships was created, DAX measures were defined, and interactive visuals were arranged into a dashboard that presents business performance for decision-making. Advanced NoSQL querying with the MongoDB aggregation pipeline complemented the relational analysis.

Illustrative Examples of Concepts Practised

The following short examples illustrate the kinds of queries and code used while learning the above modules.

```
-- Relational design (normalized tables, DDL)
CREATE TABLE customers (
  customer_id SERIAL PRIMARY KEY,
  name        VARCHAR(100) NOT NULL,
  email       VARCHAR(120) UNIQUE
);
CREATE TABLE orders (
  order_id    SERIAL PRIMARY KEY,
  customer_id INT REFERENCES customers(customer_id),
  order_date  DATE,
  amount     NUMERIC(10,2)
);
```

```
-- Join with aggregation (GROUP BY / HAVING)
SELECT c.name, COUNT(o.order_id) AS total_orders,
       SUM(o.amount) AS total_spent
FROM customers c
JOIN orders o ON o.customer_id = c.customer_id
GROUP BY c.name
HAVING SUM(o.amount) > 10000
ORDER BY total_spent DESC;
```

```
-- Performance: index + query plan
```

```
CREATE INDEX idx_orders_customer ON orders(customer id);
EXPLAIN ANALYZE
SELECT * FROM orders WHERE customer id = 42;

-- Security: least-privilege role
CREATE ROLE analyst LOGIN PASSWORD '*****';
GRANT SELECT ON customers, orders TO analyst;
```

```
// Node.js + Express: parameterized query (prevents SQL injection)
app.get('/customers/:id', async (req, res) => {
  const result = await pool.query(
    'SELECT * FROM customers WHERE customer id = $1',
    [req.params.id]
  );
  res.json(result.rows);
});
```

```
// MongoDB aggregation pipeline
db.orders.aggregate([
  { $match: { status: "delivered" } },
  { $group: { id: "$customer id", total: { $sum: "$amount" } } },
  { $sort: { total: -1 } },
  { $limit: 5 }
]);
```

4.9 Expected Outcomes

Learning Outcomes

- Designed normalized relational databases using ER modeling and created and manipulated them with SQL.
- Wrote advanced SQL queries using joins, subqueries and aggregation, and worked with MongoDB documents and aggregation pipelines.
- Applied ETL and data transformation techniques and built dashboards with data models, relationships and DAX in Power BI.
- Integrated databases with Node.js/Express and FastAPI/SQLAlchemy applications through RESTful CRUD APIs.
- Applied indexing, query optimization, access control, SQL injection prevention and backup strategies to improve performance and security.
- Understood scalability and high-availability concepts and reviewed real-world case studies from e-commerce, social media and financial services.
- Developed an end-to-end, data-driven business analytics dashboard as the capstone project.

Conclusion

The internship provided a complete and practical view of database management systems, from designing a relational schema to delivering business insights through a dashboard. By working with both SQL and NoSQL databases, integrating them with

backend APIs, and studying performance, security and scalability, the internship strengthened both the technical and professional skills required for data-centric software roles.

4.10 Certificates of Completion and Communication Proof

The AICTE – EduSkills Internship Certificate (attached before Chapter 4) was issued to Prateek Singh on Sep 10, 2026 (Certificate ID: 2026-D6F6DF349A) for completing the 8-week program in the domain of SQL and Database Management Systems. The Internship Offer Letter received from EduSkills is attached below as communication proof.

Offer Letter Ref. No.: C16Y2026M051441098

Student: Prateek Singh, B.Tech (Computer Science and Engineering), IILM University, Greater Noida

Internship Domain: SQL and Database Management Systems

Internship Duration: April 2026 – June 2026

AICTE Student ID: STU68346ab0a07001748265648

Internship Offer Letter (Page 1)



Ref No: C16Y2026M051441098

Internship Offer Letter

Student Name : Prateek Singh
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : SQL and Database Management Systems(Hindi)
Internship Duration : April 2026 - June 2026
AICTE Student ID : STU68346ab0a07001748265648

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Internship Offer Letter (Page 2 – Week-wise Structure)

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

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Page 1



The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Database Foundations & Relational Design	Introduction to database concepts, SQL vs NoSQL systems, environment setup, relational database design, ER modeling, normalization, DDL and basic DML operations (SELECT, INSERT, UPDATE, DELETE).
Week 2	Advanced SQL & NoSQL Fundamentals (MongoDB)	Advanced SQL queries including joins, subqueries, aggregation, GROUP BY/HAVING, introduction to NoSQL concepts, MongoDB document model, and CRUD operations.
Week 3	Data Aggregation, ETL & Power BI Fundamentals	Advanced SQL aggregation, ETL concepts, data transformation techniques, introduction to Power BI interface, data connections, and building basic dashboards and visualizations.
Week 4	Data Modeling & Full-Stack Database Integration (Node.js)	Power BI model view, relationships, basic DAX, performance optimization, integrating databases with full-stack applications using Node.js & Express, PostgreSQL connection, and CRUD API development.
Week 5	Backend APIs & Data Integration (FastAPI)	FastAPI with SQLAlchemy, building RESTful CRUD APIs, API consumption, using requests library, storing API data, and combining multiple datasets.
Week 6	Database Performance, Security & Backup	Performance metrics, indexing strategies, query optimization, configuration tuning, user roles & permissions, SQL injection prevention, backup and recovery strategies.
Week 7	Scalability, High Availability & Real-World Use Cases	Scaling strategies, replication, sharding, clustering concepts, case studies from e-commerce, social media, financial services, and consolidated best practices.
Week 8	Advanced NoSQL & Time Series Modeling & Capstone Project	MongoDB aggregation pipeline, advanced indexing, geospatial queries, time series data modeling, data quality, reproducibility, model validation, and deployment concepts.
Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.



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